Chapter 10

Software and Freedom

Until the 1980s, the development of computers and software was marked by collaboration and shared contributions. During this period of rapid technological growth, commercial interests were relatively minimal. However, in 1979, a popular word processing software called Scribe introduced a hidden, secret code in its program, marking the beginning of a new era. This software was designed with a "time bomb"—a piece of code that would render the program inoperable exactly 90 days after being copied and run on a system. The only way to reactivate it was for the user to pay the manufacturing company.

This incident caused a major uproar in the software development world at the time. Renowned computer programmer Richard Stallman famously criticized this practice, stating:

"This is the murder of programmers' ideals. Instead of upholding the principles of sharing and collaboration, we are now forced to bow to corporate demands—creating a system where knowledge must be paid for."

Richard Stallman

But times have changed. Business has become heavily monopolized, with profit-driven motives now dominating the field. Nevertheless, glimmers of hope remain—let us examine how.

Software and its Program

To create software, there are many programming languages like Python, C, Java, Rust, and more. We have already learned how to write small programs in Python.

The human - readable instructions (written in a programming language) that make up a program are called source code (see Figure 10.1). This source code can be edited based on the syntax rules of the programming language.

Before a computer can run a program, the source code is converted into a format that the machine can process. This converted form is called object code (or executable code). Unlike source code, object code cannot be easily modified.

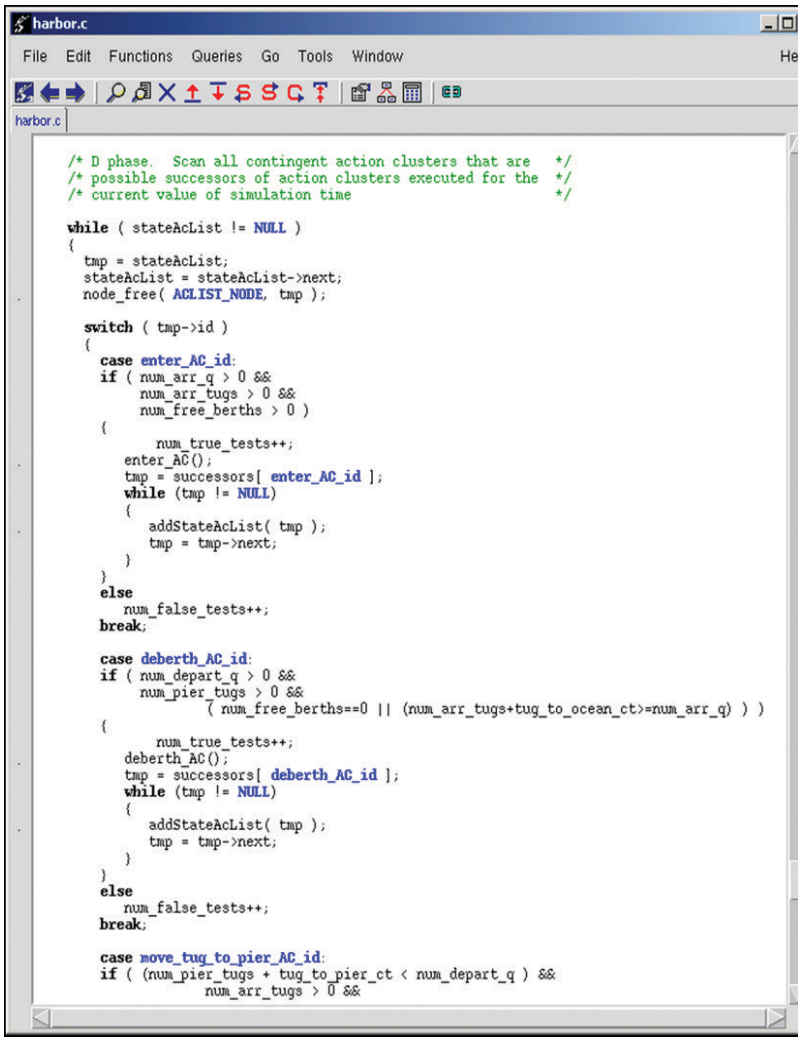


Fig. 10.1 Source Code of a Python Program

Python and Source Code

The creators of the Python programming language included a key principle in its foundational guidelines (The Zen of Python): "Code should be written in a readable and clear manner." This is why Python programs are typically distributed as source code itself. When the program is executed using an interpreter, the source code is converted into object code on the fly.

Since programs distributed as source code it allows users to read, learn, and modify them, the software's license terms must explicitly specify the conditions under which changes to the source code are permitted.

However, there is another side to this. Suppose the source codes of software are made available for others to examine and study. What are the outcomes?

- Users can examine how the software works, allowing them to trust and confidently use the programs they rely on.
- Those who are interested can learn the program. Those who are skilled can make changes and improve it if necessary. Improved or unimproved versions can be distributed. This will allow knowledge to reach more people.
- Collaborative development leads to higher-quality software. The likelihood of bugs and vulnerabilities is reduced. This reduces the risk of computer viruses and malware.
- Software companies move away from sales monopolies and instead focus on efficiency and quality service.
- Since the software is not proprietary, companies using it can produce tools at lower costs, making technology more affordable.
- .....

Let's Check the Source Code

Keeping the above points in mind, let's analyze the source code of a small program:

The software Calculator.pyc written in Python, is given in the School Resources/Class_10 folder on your computer. Try running this file using python 3.

Observe the current button colours in the software (Figure 10.2). What if you want to change these colours to your preferred shades?

To Run the File Using Python3

- Open the terminal by right-clicking on the folder where the Python file is saved and selecting Open in Terminal.
- Type the command below in the terminal and press Enter.
  python3 filename (Ex: python3 Calculator.pyc)

For this, we need to make the necessary changes to the Calculator software program and update it. Since the file Calculator.pyc is the object code of the software, we cannot edit it. So, if we want to change the program, we need its source code. This is provided in the Source folder as a Python file called calculator.py .

Open this file, find the code that gives the colour to the buttons, change it to a different colour, and try running it.

What if you want to change the calculator's name from Tom to yours?

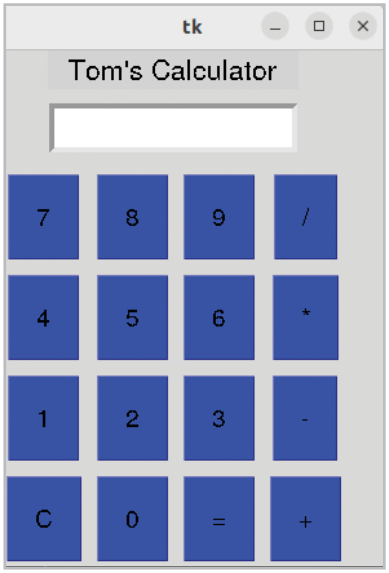


Fig. :10.2 Calculator Software Prepared in Python

Software Publishing

Creating software is just the first step—it must also be published effectively for others to use. When releasing software, keep these points in mind:

1. The software should be made publicly available in a way that allows easy download and use.
2. If users are permitted to read, modify, and redistribute the software, its source code must also be published. The terms for modification and redistribution should be clearly stated in an easy-to-understand manner.
3. Allow the software to be improved (upgraded) or forked (modified into new versions). Different versions should be organized sequentially for clarity.
4. Clearly define the license conditions for any modified versions of the software.

There are several services available that incorporate all the above possibilities and help publish software online. Below are some of them:

- GitLab (https://about.gitlab.com)
- GNU Savannah (https://savannah.gnu.org)
- Gitea (https://about.gitea.com)
- GitHub (https://github.com)
- SourceForge (https://sourceforge.net)

When publishing the developed software, the terms and conditions we specify must be formalized into a legally valid one. Since it is difficult to legally register the terms when we prepare them ourselves, we can opt for pre-existing standardized licenses prepared by established organizations.

A software license defines the rights granted to users regarding how the software may be used. Below are some software licenses:

- GNU Public Licence
- BSD (Berkeley Software Distribution) licence
- End-User Licence Agreement (EULA)

These all provide different usage rights to the user. See Table 10.1 below.

	Rights granted	Various licenses		
		GPL	**Apache**	**EULA**
0	for the user to use for any purpose.	provides	provides	No
1	reading and learning source code	provides	provides	No
	copying and editing as needed	provides	provides	No
2	To distribute copies of the software and to provide follow-up services using them	provides	provides	No
3	To redistribute modified or unmodified versions	Provides on condition	provides	No
A	To obtain another license for the modified parts	No	provides	No
B	To distribute together with other software with different licenses (Software package)	No	provides	No

Table 10.1 License Usage Rights

Beyond the previously mentioned licenses, there exist other licenses that either grant additional rights or impose further restrictions. See the following examples.

- The source code can be distributed with or without modifications. However, you must ensure that the original license applies to any modifications or additions. Such software can be called free software. These can only be used by combining them with other free software, such as the GPL (General Public License) published by GNU. GIMP is software distributed under this license.
- Software can be modified and redistributed, and the programmer can decide the terms of distribution of (only) the combined parts. There is no problem distributing it as a single package with other non-free software. These are generally called open source licenses. There are several open source licenses that vary in detail. The Apache License, which distributes the Android operating system, is an open source license. The Berkeley Software Distribution (BSD) License is an example.
- As mentioned, there are software that only provides object code. Some of them can be installed on any system. Some can be used indefinitely. Some software is also distributed for free. These fall into the category of freeware
- Some of the above types can only be used for a limited time (Testware). These do not grant any other rights.
- Software that provides only object code will have usage restrictions, allowing it to run only on a registered, specific system. Even if you have another computer, a new license must be obtained for it. Operating systems like Microsoft Windows and Apple iOS fall under this category.

In essence, free software prioritizes principles such as user freedom, the dissemination of knowledge, ethics, justice, and public good. Most free software is developed through voluntary efforts and is available free of charge.

Open-source software, on the other hand, emphasizes universal accessibility, collaboration, usability, improvement, and innovation. A middle-ground approach has emerged in the form of FLOSS (Free/Libre and Open Source Software) licenses.

(The English word 'Free' in this context refers to 'freedom' rather than just 'zero cost.' Such licenses emphasize that the software must be free—as in liberty—not necessarily free of charge.)

Versions and Forks

Any software in use undergoes periodic updates over time. The purpose of creating a new version is to rewrite the security code of the program, improve existing features, add new functionalities, etc. Software versions are typically indicated by a numbering system. For example, Kazium Version 21.12.3 means the 21st major version of the Kazium software, the 12th interim improvement, and the third minor changes (patches).

Fork is applicable only in open/free software. Those who disagree with the views, methods, or features of existing software - rewrite and distribute the software source code as permitted by the software's license. For example, the two forks we use, LibreOffice (published by The Document Foundation) and Apache OpenOffice (published by the Apache Software Foundation), are two forks of the defunct OpenOffice software.

Let's Check the Licenses

As mentioned earlier, the GNU General Public License (GPL) falls under the first category of free software licenses. The key terms of this license (listed as items 0, 1, 2, and 3 in Table 10.1) should be noted separately in your notebook. These terms define the essential conditions for free software.

Why is the First Term Numbered '0'?

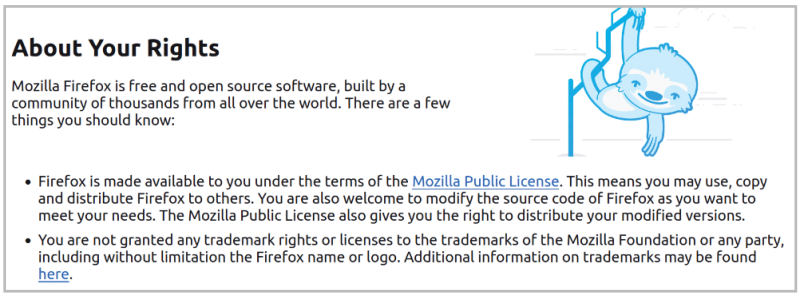
Check Software Licenses on your Computer as given below.

To Check Licenses

To check the license of the Krita software on our operating system,

- Open the software window and check the Help menu in the menu bar.
- When you click About Krita, check the License item in the window that appears.
- It should be noted that Krita is distributed under the GPL license.

Mozilla Firefox, the web browser we use on our computers, is licensed under the Mozilla Public License. This is a FLOSS license. See Figure 10.3 below.



About Your Rights

Mozilla Firefox is free and open source software, built by a community of thousands from all over the world. There are a few things you should know:

- Firefox is made available to you under the terms of the Mozilla Public License. This means you may use, copy and distribute Firefox to others. You are also welcome to modify the source code of Firefox as you want to meet your needs. The Mozilla Public License also gives you the right to distribute your modified versions.
- You are not granted any trademark rights or licenses to the trademarks of the Mozilla Foundation or any party, including without limitation the Firefox name or logo. Additional information on trademarks may be found here.

Fig. 10.3 License of Mozilla Firefox

Complete Table 10.2 by identifying the licenses for the software we use for various purposes. Check on the internet if needed.

Software	License	Category	Special Terms
LibreOffice	Mozilla Public Licence	FLOSS	Trademarks used in the software may not be freely used.
Scratch			
Firefox			
Chromium			
GeoGebra			

Table 10.2 Details of Software Licenses

History of Licenses

As we know, progress in the world has always been built upon prior knowledge and discoveries. By the 1980s, a debate emerged that software source code—like literary works such as stories and poems—should be recognized as original intellectual creations. This idea gained legal acceptance first in the United States and later in other countries, bringing program source code under

copyright law. As a result, reusing a program's code or parts without permission became legally equivalent to plagiarism or literary theft.

A programmer's work must be protected from being stolen by others. This is society's responsibility towards that individual. However, this led to the hiding of program codes and attempts to establish monopolies in software. To address this, software licenses came into use, some of which strongly oppose the free dissemination of knowledge..

Free Software Foundation

Along with the aforementioned commercialization, there were indeed strong counter-movements. One such pioneering initiative was the GNU Project, established in 1983 by renowned computer programmer Richard Stallman. The primary objective of this project was to create and make freely available software that could be used with complete freedom. To support these efforts, the Free Software Foundation (FSF) was formed to create and improve software for the GNU Project, a non-profit organization of volunteer programmers that began in 1985. Today, the organization has branches and many activists around the world.

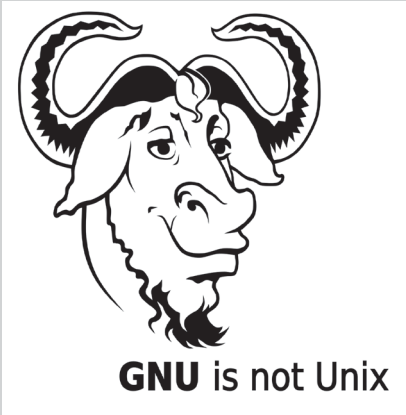
Unix Operating System

Developed during 1969–70 under the leadership of Ken Thompson, Dennis Ritchie, and Douglas McIlroy at AT&T (one of the world's largest telecom and software companies at the time), Unix became a foundational operating system. Its development began in 1969, and later, its ownership rights were transferred to various other companies. These versions, after significant changes and additions, have been used in the operating systems of Apple Corporation (Mac OS X) and the University of California, Berkeley (BSD - Berkeley Software Distribution). Many of today's operating systems are developed from Unix or have been rebuilt based on Unix.

The GNU Project and Its Software

The GNU Project, founded by Richard Stallman, attracted programmers from around the world who contributed without profit motives, creating nearly all essential free software components.

GNU (GNU is not Unix)



When Richard Stallman called on the programming community around the world to create a new operating system, hundreds of programmers came forward to collaborate with the GNU Project. Most of them worked for various other organizations. But Stallman was unable to bring them together or explain the project at the time.

With these constraints, the practical approach was to create a new system modeled after an existing one. Therefore, Stallman asked for the Unix operating system to be used as a model in this regard. Everything that was in Unix should be included in the new operating system. But it was imperative that none of the Unix program codes be used. Then, the programmers who collaborated on the project rewrote Unix, without copying the codes. This is what the name GNU (GNU is not Unix) refers to.

Copy Left

If a software developer wishes to grant users the freedom to use, copy, modify, and redistribute a program, these rights must be legally guaranteed. Copyleft is a licensing method that uses copyright law itself to ensure that all users retain these freedoms.

Copyleft does not mean that the software is unregistered or unprotected by copyright law. Rather, it means that the rights have been formally registered and granted to the user under specific terms that ensure freedom.

The GNU Project and Linux

An operating system has two main parts at the functional level. One part is called the Kernel and the other part is called the Shell. The kernel is the part that directly interacts with the computer's hardware. The shell is the part that interacts with the user. The graphical user interface (GUI) part of the operating system, which consists of software windows, is associated with the shell (Figure 10.4).

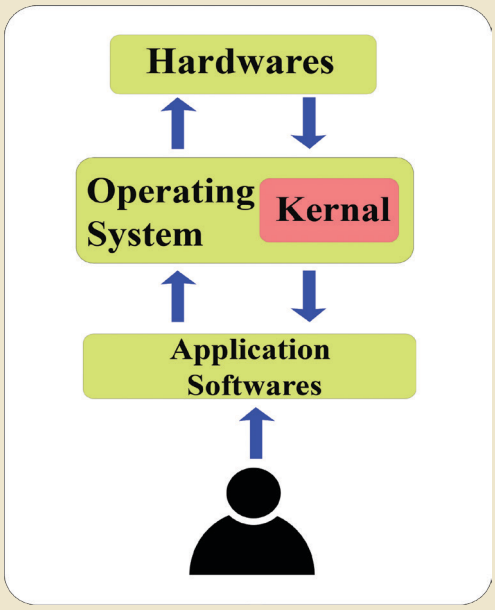


Fig.:10.4 Operating system – working phase

to build was called The Hurd. However, even after all the other software of GNU was completed, the Hurd was not completed. Around the same time, a Finnish programmer named Linus Torvalds completed and published a kernel program called Linux. Torvalds, who had listened to Stallman's requests for GNU, built the new kernel in such a way that it could run the software that the GNU Project was developing. The combination of these two was the GNU/Linux operating system that we use today.

The kernel and other parts of the GNU/Linux operating system we currently use are given below.

Kernel	Linux
Shell	GNU Bash (GNU Bourne Again Shell)

The term "Shell" originates from the Unix operating system. When examining how GNU developed its version, the name "Bash (GNU Bourne Again Shell)" is highly meaningful.

Free Software Development

You may have heard of the Microsoft Windows operating system. Microsoft, a proprietary software company, employs its own programmers to develop and improve Windows.

However, free and open-source software (FOSS) is not developed this way. Instead, it follows a different approach, as described below.

- Publishing the concept, methods, and features of a software on the Internet.
- A prototype of it is being published.
- Any user from any corner of the world can participate in the subsequent activities. They can discuss ideas, add program codes and can correct mistakes.

What Infrastructure is Needed for This Approach?

- Many contributors will create multiple versions of the software. These versions must be properly stored and managed.
- Each version should indicate which parts of the source code have been modified and which changes have been proposed.
- If major changes cause issues, developers should be able to roll back to a previous version. Hence, every version must be preserved.
- .....

Git is an online software that handles all of the above needs. It is a system created by Linus Torvalds, the creator of the Linux kernel (Figure 10.5), for the collaborative improvement of his software. Apache Subversion, developed by the Apache Software Foundation, is a software for centralized improvement.

GitHub, GitLab, GNU Savannah are popular platforms that rely on such version control software for publishing and managing code.

GNU-Linux Distributions

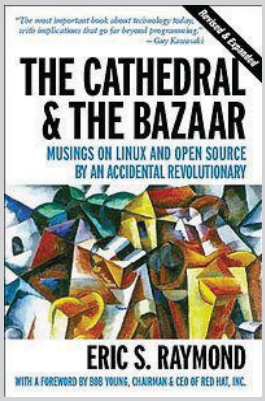
We have discussed that GNU-Linux is licensed in a way that allows anyone to copy, modify, and redistribute it. Taking advantage of this opportunity many versions of GNU-Linux programs have been created. These distributions vary widely based on the developers' vision and the specific needs they aim to fulfill.

Fig. 10.5 Linus Torvalds

Cathedral Model and Bazaar Model

Have you noticed the ways Microsoft and Linux build software?

Eric Raymond, a famous writer on technology, called this the Cathedral Model and the Bazaar Model.



Some GNU-Linux Distributions

- Debian GNU/Linux (from which there are many versions, such as Ubuntu and the Kite GNU-Linux we use).
- Red Hat Linux
- Slackware Linux

(There are many other versions of each of these.)

Android

Android is a widely used operating system in touchscreen mobile phones, tablets, and other devices. It is based on the Linux kernel, and its program code is available under an open-source license.

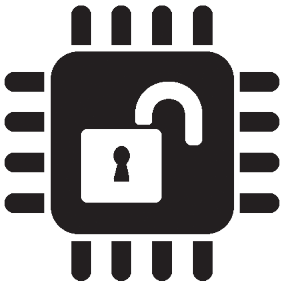
Android is a prime example of how free and open-source software (FOSS) benefits society. By using Android, manufacturers can produce smart devices like mobile phones and sell them at lower prices, as they don't have to invest heavily in developing their own operating systems.

Open Hardware

It's not just software that is being monopolized. The technology behind computers, other electronic devices, and mechanical components is also being monopolized in this way.

These are the following stages in the manufacturing of an electronic device.

- Outline of the general organization of machine , illustrations, diagrams, etc. that help in the construction of components
- Required component structures and their manufacturing methods
- Layout diagrams of electronic chips
- Special software (drivers) for the operation of components
- .....



If the above-mentioned [technologies/components] are obtained, any company with the technical capabilities to manufacture electronic devices can produce them.

Conventionally, large corporations invest significant capital in research and development to create specialized equipment, but they rarely publish their manufacturing methods.

However, just as we saw earlier with free and open-source software, there is now a growing open-design movement for hardware components released under free manufacturing licenses.

An example of this is the Arduino microcontrollers (Figure 10.6), which are widely used in schools and follow open-hardware principles.

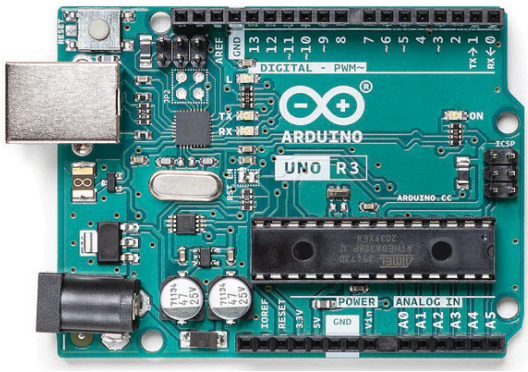


Fig. 10.6 Arduino Uno board

Let's Evaluate

- Which of the following is not an open source software license?
  a) Apache Software License
  b) End User License Agreement
  c) GNU General Public License
- Which of the following is free hardware?
  a) Apple mobile phones
  b) Arduino similar chipset
  c) Mobile phones with Android installed
- What is the name given to free software?
  a) Free software
  b) Open software
  c) Freeware
- What is the shell in the GNU/Linux operating system known as?
  a) Open Shell
  b) Linux Shell
  c) Bash
- Which of the following is not an operating system?
  a) Android
  b) Linux
  c) Unix

Extended Activities

1. Complete the table below by searching familiar operating systems on the internet.

Operating system	Based program	The device used
Ubuntu	Linux	Computer
iOS	Unix	
Android	Linux	mobile phone,.....

2. Search the internet for the names of operating systems used in wearable devices such as smartwatches and VR headsets, and make a list.